Harmonic fit of Migkas $A(\ell)$ set equal to the Sedov–Taylor + Gauss tidal model

Notation. r: radius of our vision shell (horizon), centred on us, $r \simeq 14.3$ Gpc $\simeq 46.5$ Gly comoving. D: our distance from the CoE, beyond the horizon, so $d < r < D$. d: source (cluster) depth from us. R: distance from the CoE; R_S for a source. L: density scale length. $\kappa \equiv GM(<D)/(H_0^2 D^3) = M(<D)/[2M_{\rm crit}(<D)]$ (definition used in the numerical model; uniform Ω_m gives $\kappa = \Omega_m/2$).

Model. Let $\vec{D}$ point from the CoE to us, so $\hat{D}$ is the anti-CoE axis (ℓ_0, b_0), and let $\mu = \hat{n}\cdot\hat{D} = \cos\theta$ for line of sight $\hat{n}$. A source at $\vec{R}_S = \vec{D} + d\,\hat{n}$ has (law of cosines) $R_S^2 = D^2 + d^2 + 2Dd\,\mu$. For a point-mass $(1/R^2)$ field the peculiar velocity is $\vec{u} = -k\vec{R}/R^3$, and $\delta H = (\vec{u}_S - \vec{u}_O)\cdot\hat{n}/d = (k/D^3)\left[(3\mu^2-1) + \frac{d}{D}\left(\frac{9}{2}\mu - \frac{15}{2}\mu^3\right) + O(d^2/D^2)\right]$ (checked numerically: next-order coefficients 4.5000, -7.5000; for $d/D < 0.022$ the neglected terms are $\sim 5\times10^{-4}$ of the leading term), hence

$$\frac{A}{A_{\rm all}} = N\left[1 + \alpha\kappa\,(3\mu^2-1) + \alpha\kappa\,\frac{d}{D}\left(\tfrac{9}{2}\mu - \tfrac{15}{2}\mu^3\right)\right]. \tag{1}$$

In the Galactic plane, $\mu = c\cos x$ with $c \equiv \cos b_0$, $x \equiv \ell - \ell_0$. Using $\cos^2 x = \frac{1}{2}(1+\cos 2x)$ and $\cos^3 x = \frac{1}{4}(3\cos x + \cos 3x)$, and the 90° sector window $W_m = \dfrac{\sin(m\pi/4)}{m\pi/4}$ ($W_1 = 0.900$, $W_2 = 0.637$, $W_3 = 0.300$):

$$\begin{aligned}\frac{A}{A_{\rm all}} = {} & N\left[1 + \alpha\kappa\left(\tfrac{3}{2}c^2 - 1\right)\right] + N\alpha\kappa\,\frac{d}{D}\,c\left(\tfrac{9}{2} - \tfrac{45}{8}c^2\right)W_1\cos x \\ & + N\alpha\kappa\,\tfrac{3}{2}c^2\,W_2\cos 2x - N\alpha\kappa\,\frac{d}{D}\,\tfrac{15}{8}c^3\,W_3\cos 3x.\end{aligned} \tag{2}$$

Data. Linear least squares to the 71 digitized points of Migkas et al. (2020), Fig. 6:

$$\frac{A}{A_{\rm all}} = 0.9965 + 0.0296\cos\ell + 0.0434\sin\ell + 0.0636\cos 2\ell + 0.0366\sin 2\ell - 0.0299\cos 3\ell - 0.0114\sin 3\ell, \tag{3}$$

$R^2 = 0.915$ (dipole+quadrupole only: $R^2 = 0.814$). Quadrupole phase: $\ell_2 = 15.0°$.

Term-by-term equality (data coefficient of $\cos m(\ell - 15°)$):

$$m=0: \quad 0.9965 = N\left[1 + \alpha\kappa\left(\tfrac{3}{2}c^2 - 1\right)\right] \tag{4}$$

$$m=1: \quad +0.0398 = N\alpha\kappa\,\frac{d}{D}\,c\left(\tfrac{9}{2} - \tfrac{45}{8}c^2\right)W_1 \tag{5}$$

$$m=2: \quad +0.0733 = N\alpha\kappa\,\tfrac{3}{2}c^2\,W_2 \tag{6}$$

$$m=3: \quad -0.0292 = -N\alpha\kappa\,\frac{d}{D}\,\tfrac{15}{8}c^3\,W_3 \tag{7}$$

Manipulations.

$$\frac{m=1}{m=3}: \quad \frac{\left(\frac{9}{2} - \frac{45}{8}c^2\right)W_1}{-\frac{15}{8}c^2 W_3} = \frac{0.0398}{-0.0292} \;\Rightarrow\; c^2 = 0.695,\ |b_0| = 33.5° \tag{8}$$

$$\frac{m=3}{m=2}: \quad \frac{0.0292}{0.0733} = \tfrac{5}{4}\,c\,\frac{W_3}{W_2}\,\frac{d}{D} \;\Rightarrow\; \frac{d}{D} = 0.81 \tag{9}$$

$$m=2: \quad N\alpha\kappa = \frac{0.0733}{\frac{3}{2}c^2 W_2} = 0.1105 \tag{10}$$

$$m=0: \quad N = 0.9918, \qquad \alpha\kappa = 0.111 \tag{11}$$

Formal solution. $\ell_0 = 15.0°$, $|b_0| = 33.5°$, fitted "d/D" $= 0.81$, $\alpha\kappa = 0.111$, $N = 0.992$; all four equations are satisfied exactly; $R^2 = 0.780$ against the 71 points. The components perpendicular to the axis (coefficients of $\sin m(\ell - 15°)$: $+0.0342$, -0.0001, $+0.0131$) cannot be produced by any axisymmetric model; the dipole direction $\ell_1 = 56°$ is 41° off axis.

Physical interpretation for $d < r < D$. Median Migkas redshift $z = 0.075$ (Migkas et al. 2020, Sect. 2), $H_0 = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$:

$$d \simeq \frac{cz}{H_0} = \frac{(2.998\times10^5)(0.075)}{70}\ \mathrm{Mpc} = 321\ \mathrm{Mpc} \approx 1.05\ \mathrm{Gly}, \qquad r \simeq 14.3\ \mathrm{Gpc} \approx 46.5\ \mathrm{Gly}, \tag{12}$$

$$D > r \ \Rightarrow\ \frac{d}{D} < \frac{d}{r} = \frac{0.321}{14.3} = 0.022 \qquad (\text{and } d/D < r/D < 1 \text{ for any source inside the shell}). \tag{13}$$

Putting $d/D = 0.022$ into Eqs. (5) and (7), with $N\alpha\kappa$ from Eq. (6), for $|b_0| = 33.5°,\ 0°,\ 15°$:

$$m = 3: \quad -N\alpha\kappa\,\tfrac{d}{D}\,\tfrac{15}{8}c^3W_3 = -0.00081,\ -0.00097,\ -0.00094 \quad [\text{meas. } -0.0292], \tag{14}$$

$$m = 1: \quad N\alpha\kappa\,\tfrac{d}{D}\,c\left(\tfrac{9}{2} - \tfrac{45}{8}c^2\right)W_1 = +0.0011,\ -0.0017,\ -0.0012 \quad [\text{meas. } +0.0398]. \tag{15}$$

The predicted octupole is 30–36 times too small, and the predicted on-axis dipole is 23–36 times too small (with the wrong sign for $|b_0| \le 15°$). **The fitted $d/D \approx 0.8$ therefore cannot be the literal cluster-depth-to-CoE ratio.** It is an effective shape parameter $\varepsilon_{\rm eff}$ that sets the octupole-to-quadrupole ratio, $a_3/a_2 = \frac{5}{4}c\,(W_3/W_2)\,\varepsilon_{\rm eff}$ [Eq. (9)]. (Taken literally, $\varepsilon_{\rm eff} = 0.76$–$0.81$ would give $D = d/\varepsilon_{\rm eff} \approx 400$ Mpc $\ll r$, with $M(<D) = 2\kappa M_{\rm crit}(<D) \approx (3\text{–}4)\times10^{19}M_\odot$ and $L \approx 52$–56 Mpc. That puts the CoE inside the shell, contradicts $d < r < D$, and is not used.)

What the fit fixes for $d < r < D$. As $d/D \to 0$ only $m = 0, 2$ survive, so the CoE part of the signal is the in-plane quadrupole: axis $\ell_0 = 15.0°$ (mod 180°); tidal amplitude $N\alpha\kappa\cos^2 b_0 = a_2/(\frac{3}{2}W_2) = 0.077$, i.e. $\alpha\kappa = 0.080$ ($b_0 = 0°$) to 0.085 ($|b_0| = 15°$). The $\alpha\kappa = 0.111$ of Eq. (11) belongs to the $|b_0| = 33.5°$ solve that uses $\varepsilon_{\rm eff}$. $|b_0| \lesssim 15°$ comes from the full-model fits (which used $d/D \approx 0.76$). The quadrupole alone does not constrain b_0 (it is degenerate with $\alpha\kappa$). Because $3\mu^2 - 1$ is even in μ, it does not tell the CoE side ($\ell \approx 195°$) from the anti-CoE side ($\ell \approx 15°$); that assignment comes from the odd terms, which this geometry does not produce at the observed level. A pure axisymmetric quadrupole explains $R^2 \le 0.53$ of the 71 points (the quadrupole-only LSQ gives 0.532).

Full numerical model. `dq_model_lib.model_fine` and `make_model_param_heatmaps.model_A` put us at $D\hat{x}$ from the CoE and the sources at $D\hat{x} + d\,\hat{n}$ with $d = (d/D)\,D$: the same literal meaning. They include the exponential-core profile $\rho(R) = \rho_c e^{-R/L} + \rho_u$ (R from the CoE; floor fraction $f_u = 0.94$; $\rho(D) = 0.3\rho_{\rm crit}$ before the fit rescales the mass by $0.458/0.385 = 1.19$, i.e. $\approx 0.36\rho_{\rm crit}$) and the time-integrated drag history. At $d/D = 0.022$, scanning $D/L = 3$–20, $b_0 = 0°, 15°$ and $f_u = 0$–0.99 (κ only scales the amplitude) gives $|a_3/a_2| \le 0.012$ (0.024 without the 90° window) and a_1/a_2 from -0.022 to $+0.037$, against measured $a_3/a_2 = -0.40$ and $a_1/a_2 = +0.54$. The profile and history change the octupole coefficient by a factor $\lesssim 2$, not ~ 30. **No parameter of the model reproduces a_3 with $d/D \le 0.022$.** Even at $d/D = 0.76$ the grid reaches only $|a_3/a_2| \le 0.36$. An octupole this large needs a field that varies on the scale of d itself, e.g. local ($\lesssim 1$ Gpc) structure or bulk flows, depth-dependent selection or scaling-relation systematics, or sky-coverage and sector-window effects. A smooth field from a mass beyond the horizon does not vary on that scale.

Caveats. (i) Full numerical model, re-optimized: $R^2 = 0.776$ ($b_0 = 0°$), 0.777 ($15°$), 0.752 ($33.5°$), so $|b_0| = 33.5°$ is a truncation artifact and $b_0 = 0°$–$15°$ is not resolved. At $b_0 = 0°$: $\kappa \approx 0.45$–0.46, $d/D \approx 0.76$, $D/L \approx 7.6$, $\ell_0 \approx 14.4°$. At $b_0 = 15°$: $\kappa \approx 0.27$, $d/D \approx 0.74$, $D/L \approx 11.3$. All of these use $d/D \gg 0.022$, so they violate $d < r < D$ as literal geometry. (ii) $\kappa = GM(<D)/(H_0^2D^3) = M(<D)/[2M_{\rm crit}(<D)]$; $\rho_{\rm crit} = 3H_0^2/8\pi G = 1.36\times10^{11}M_\odot\,\mathrm{Mpc^{-3}}$ ($h = 0.7$). (iii) The coefficients $\frac{9}{2}, \frac{15}{2}$ are

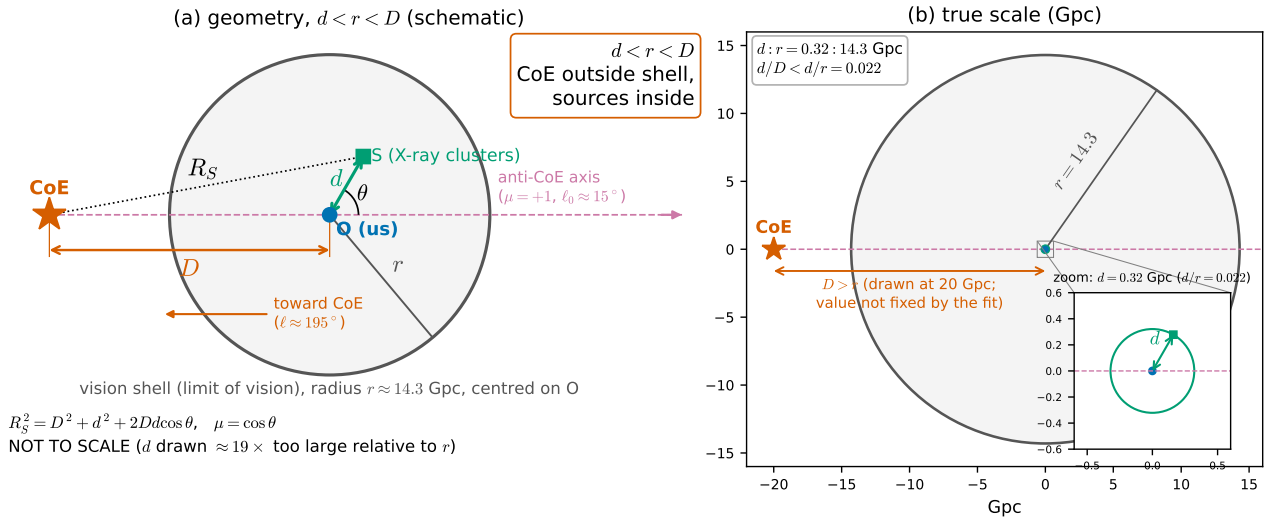


Figure 1: Geometry for $d < r < D$. (a) Schematic, not to scale: CoE beyond the vision shell (radius r, centred on us), sources S inside at depth d, $R_S^2 = D^2 + d^2 + 2Dd\cos\theta$, θ from the anti-CoE axis. (b) True scale in Gpc: $d = 0.32$ Gpc (green dot; zoom inset) against $r = 14.3$ Gpc; $D > r$ is drawn at 20 Gpc for illustration only.

point-mass values. (iv) The 71 sector points overlap and are not independent. (v) A single effective depth is used; the sample spans $z = 0.003$–0.45, i.e. $d \lesssim 1.9$ Gpc (cz/H_0), still $\ll r$.